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Hydrogen Enrichment Impact on Gas Turbine Combustion Characteristics

Bassam Abdelnabi, Ahmed Elkady, Keith Mcmanus, Anthony Brand and Daniel Cuppoletti

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Technical Report Abstract Page

Title	Hydrogen Enrichment Impact on Gas Turbine Combustion Characteristics		
Author(s)	Bassam Abdelnabi	**SSO**	200019195
	Ahmed Elkady		200013789
	Keith Mcmanus		200005585
	Anthony Brand		200003039
	Daniel Cuppoletti[1]		--
Component	Emissions & Fuel Flexibility Laboratory, Niskayuna		
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Key Words: Hydrogen enrichment, Hydorgen doping, Fuel Flexibility, NOx and CO, emissions, Gas Turbine combustion

Abstract: The current research provides the impact of H2 enrichment on gas turbine combustion characteristics while isolating parameters that influence mixing performance. Aerodynamic effects are isolated by using a perforated plate burner to provide simple flame structure (as compared to swirling flow). Also, the study is conducted under perfectly premixed conditions to isolate the effect of unmixedness. Hydrogen enrichment is varied from 0 to 25 percent by volume. The experiments are carried out over a wide flame temperature range that extends from the LBO limit up to extreme temperatures. NOx, CO emissions, LBO limits as well as flame luminosity are reported with a combustor pressure of 185 psi.

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[1] Prioir intern at GE

1 INTRODUCTION

In recent decades, gas turbine combustor development has been heavily influenced by increasingly stringent emissions requirements. Conventional (non-premixed) combustors have wide flex fuel capability, but emissions are a major challenge. The development of the Dry Low Emissions (DLE) and Dry Low NOx (DLN) technologies enabled low NOx emissions, but also added the risk of flashback, flame holding and combustion dynamics. In addition, flexible fuel operation becomes a challenge as a change in fuel propertied may result in a change in the operational characteristics of the combustor. It may be a favorable or an adverse change, but it needs to be studied carefully and quantified. This research is focused on the impact of H2% on emissions (NOx and CO) and LBO.

The impact of H2 addition on NOx emissions appears to be contradictory in published literature. Griebel et. al reported similar NOx emissions with and without H2 doping (also known as addition or enrichment) for perfectly premixed fuel air mixture [1]. However, they showed that H2 piloted operation (H2 not premixed with the fuel air mixture) significantly increase the NOx concentrations. Han S. Kim et. al [2] reported an increase in NOx with H2 addition in premixed mode. The experimental setup of Han S. Kim et. al doesn't appear to yield a perfectly premixed air/fuel mixture. Cozzi F. reported NOx increase with H2 addition for non-premixed mode of operation [3].

With regards to impact on Lean Blow Out (LBO), Hydrogen addition to natural gas fired combustors was shown to extend the LBO limit [4-6]. Flame imaging studies shows that the chemical as well as the physical characteristics of the flame is strongly influenced with H2 addition [4, 6-8]. This was demonstrated in terms of the change of the location of the heat release zone due to increased reactivity. Detailed velocity field investigations showed that with H2 addition, near LBO, the flame anchoring extends to relatively higher velocity regions (compared to no H2 addition) [4, 7-9]. However, all aforementioned studies are conducted under atmospheric conditions which provide limited insight on the actual behavior of a pressurized combustor. Dagaut and Dayma showed that, under pressure of 10 bar, OH radical concentration increases significantly with H2 enrichment and this explains the increased mixture reactivity [10]. P. Griebel et. al showed that increasing the H2 content from 0-20% (by volume) results in a linear decrease of the LBO limit [1]. Interestingly, this was reported at pressures of 5 and 14 bars and the LBO limits were almost identical with an H2 content of 15-20%, for both pressures. They also showed that significant reduction in CO is achieved with H2 enrichment, near the LBO limit.

The combustor under investigation has been extensively tested in the past with studies on emissions entitlement, impact of acoustics, kinetics modelling validation, impact of Exhaust Gas Recirculation (EGR) as well as 3D CFD validation [11-18]. This research is aimed at evaluating the effect of H2 enrichment on the combustion characteristics of a gas turbine combustor burning natural gas over a wide flame temperature range that extends from the LBO limit up to 3300 F. NOx, CO emissions, LBO limit as well as flame luminosity are reported under a pressure and a pre-heat temperature of 185 psia and 600 F respectively. Hydrogen enrichment is varied from 0 to 25% percent by volume. The study is conducted under perfectly premixed conditions to isolate the effect of unmixdness from the results.

2 NOMENCLATURE

P_3	=	Combustor inlet pressure
T_3	=	Combustor air/Fuel mixture inlet temperature
TC	=	Thermocouple
P	=	Pressure
$\Delta P/P$	=	Percentage pressure drop across Combustor
NO_x	=	Nitric Oxides
CO	=	Carbon Monoxide
O_2	=	Oxygen
CO_2	=	Carbon dioxide
H_2	=	Hydrogen
ppmvd	=	Particle per million volume dry
T_{flame}	=	Flame temperature based on O2 emissions
ms	=	millisecond
ppm/ms	=	Particle per million per millisecond
ft/s	=	Velocity in feet's/second

3 EXPERIMENTAL SETUP AND TEST CONDITIONS:

3.1 Experimental facility

The test setup is very similar to that of Elkady et. al [14]. The ASME stamped pressure vessel has a diameter of 8'' with pressure and temperature ratings of 300 psia and 1100°F, respectively. Details of the pressure vessel with air, fuel, water and exhaust connections are shown in Figure 1. Air is supplied through two independent feeds for combustion and cooling which are metered separately. High-pressure, preheated combustion air is connected to a T-connection where the fuel is injected. Prior to entering the combustor, the fuel and air mixture passes through a tube bundle, as shown in Figure 2, having a length greater than 200 tube diameters to achieve condition close to perfect mixedness. A number of thermocouples are installed on the tube surface to monitor for auto-ignition and flashback. These T/C's, along with combustor inlet temperature (T3) T/C, are connected to an interlock system that shuts down the fuel supply if auto ignition or flashback occurs (in the tube bundle or at the nozzle inlet). The cooling air is directed to the outer side of the combustor to balance the pressure across the liner. The exhaust gases emanating from the liner are mixed with the spent cooling air in the exhaust plenum where cooling water is injected to reduce the exhaust temperatures upstream of a throttle pressure control valve. The cooled exhaust gases are directed through the exhaust piping and released to the atmosphere.

The fuel used in this study is a typical pipeline natural gas having a methane content of approximately 96% doped with H2 ranging from 0-25%. Preheated 565°F is utilized to provide both combustion and liner cooling air. The total preheated air mass flow rate ranges between 1-2.4 lbm/sec depending on the operating conditions. Flame temperatures are varied from lean blow-out (LBO) up to 3300 F. Combustion air is metered using a venturi with an accuracy of better than 1%. Fuel is metered using a Coriolis type mass flow meter with an accuracy of 0.05%.

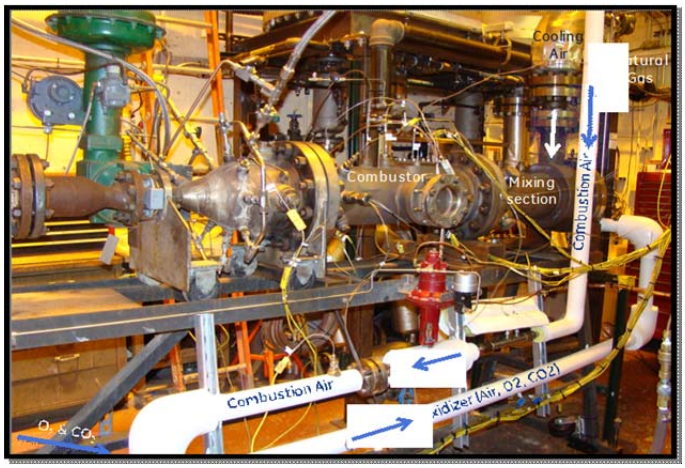


Figure 1. Pressure vessel and air/fuel connections.

Figure 2. Air /Fuel mixing tube with surface temperature monitoring

3.2 Combustor and Optical access

The combustor assembly is illustrated in Figure 3. The assembly comprises an inlet manifold with a conical entrance, burner, dome flange, quartz liner, cooling air sleeve and a mounting plate to assemble the combustor to the high pressure vessel. The air fuel mixing tube is connected to the conical entrance of the inlet manifold to straighten the reactant stream prior to a perforated plate that stabilizes the flame in the combustor. Measurements of combustor inlet pressure (P3) and inlet temperature (T3) are made at the inlet cone. Combustor is designed with a residence time of 20ms.

Details of the perforated flame stabilizer are shown in Figure 4. The stabilizer also serves as a flame arrestor, preventing flashback of the flame into the inlet manifold. The flame arrestor tubes have a diameter of 0.06 inch while the flame stabilizer tubes have a diameter of 0.125''(3.175 mm) arranged in a manner similar to the one used by Leonard et al. [1] and Elkady et al. [14]. The perforated plate is made of Inconel 718 and has 36 holes arranged as shown in Figure 4. The tube spacing is ¼'' (6.35 mm) with the center hole designed to fit a ¼ inch igniter (hydrogen torch).

3.3 Emissions Sampling

Combustion products are sampled using a water-cooled gas sampling probe placed on an axial traverse. The sample is then transferred to the sample-conditioning unit via heated line passing through a heated filter. A stream of the sample is then cooled, dried, and sent to a gas sampling system for CO_2, O_2, UHC, and CO measurement. The other stream remains heated before entering the NO_x analyzer to eliminate loss of NO_2 The gas sampling unit is calibrated

using standard calibration gases. The analyzers are spanned prior to tests and then checked during and after the experiments to ensure no drift. A recent gauge repeatability and reproducibility analysis (gauge R&R) was performed on the NO_x analyzer confirming a measurement accuracy of +/- 1 ppm of the full range. In this report, NO_x emissions are corrected and reported to a 15% O_2 level by volume.

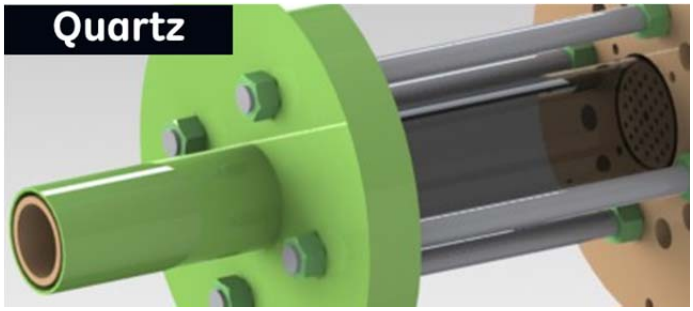


Figure 3. Combustor with Quarts liner

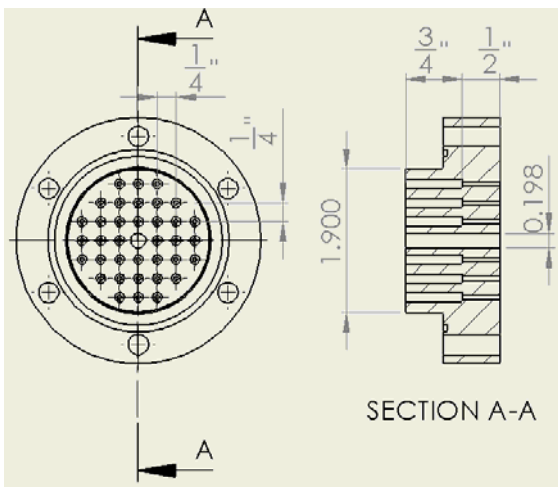


Figure 4. Perforated flame burner.

3.4 Test conditions and data acquisition

Experiments are conducted at P_3 of ~185. The combustor residence time is ~14 ms. The combustor pressure drop ($\Delta P/P3$) depends on operating conditions. The stabilizing jet velocity (calculated based on preheated air flow) varied in the range of 44-50 ft/s (13.41-15.23 m/s) depending on the test conditions.

Real-time data is acquired at a frequency of 1Hz. Reported data point represent an average of 8-10 consecutive data samples. Maximum NO_x standard deviation is roughly ±0.8ppm (1.9% mean value). The reported average P_3 and T_3 has a typical standard deviation of ±1.2 psi (0.08 bar) and ±4°F (2.2 K) respectively. Flame temperatures are calculated using fuel-air ratios based on metered air and fuel flows or the O_2 and/or CO_2 percentages in the sampled combustor products. The latter technique is valid when complete combustion occurs; i.e., at flame temperature above the LBO limits. In this study, the flame temperature based on O_2 and CO_2 typically match to within 10-15 degrees. In addition, the flame temperature based on metered air

and fuel matches with the temperature based on O2 emissions within 1%. In this paper, reported flame temperatures are calculated based on O_2 concentrations. The uncertainty in flame temperatures is estimated to be ±40°F (22.2 K).

4 RESULTS AND DISCUSSIONS

4.1 Impact of H2 doping on NOx emissions

Figure 5 shows the effect of hydrogen addition on NOx emissions corrected to 15% exhaust O2. The results demonstrate that there is minimal to no effect of H2 enrichment on the NOx levels (up to 25% addition). To provide an insight on this behavior, it is of importance to study the impact on H2 addition on Tflame.

Figure 6 shows the Tflame vs equivalence ratio for H2 blends ranging from 0 -100%. The temperatures calculated from equilibrium assumption are similar to the experimentally determined values. The flame temperatures with hydrogen addition are higher than the pure natural gas flames, but there is very little difference in flame temperature below 50% hydrogen. The difference is around 15 degrees with 25% hydrogen. However, the increase is dramatic beyond that regime.

This impact of H2 on Tflame provides possible explanation on why published data appears to be contradictory when it comes to impact on NOx emissions. For example, taking close look at the data published by Han S. Kim et. Al [2], we notice that they reported NOx increase under premixed mode, but the H2 percentage was between 40-60%. It appears that the difference is driven by the Tflame difference.

The sensitivity to flame temperature is because the dominant reaction in the thermal NOx generation mechanism (O + N2 ↔ NO + N) is dependent on flame temperature. A closer look at the Figure 5 shows that the NOx emissions are fraction of a ppm higher with Hydrogen enrichment. This is attributed to the increased OH concentration which contributes to the NOx generation through the extended Zeldovich mechanism (N + OH ↔ NO + H).

It is of prime important to maintain constant Tflame while doing such comparison. Several publications donot to take that into consideration even when studying the impact of air preheat temperature. Actually, air preheat temperature has a pronounced effect on Tflame so a comparison under the same equivalence ratio doesnot provide back to back comparison.

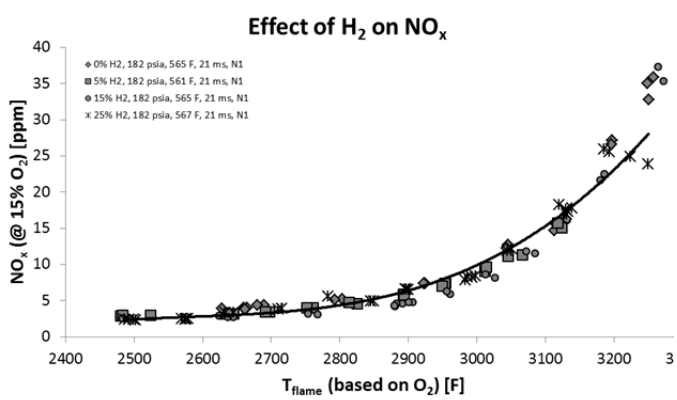


Figure 5. Impact of H2 % on NOx emissions

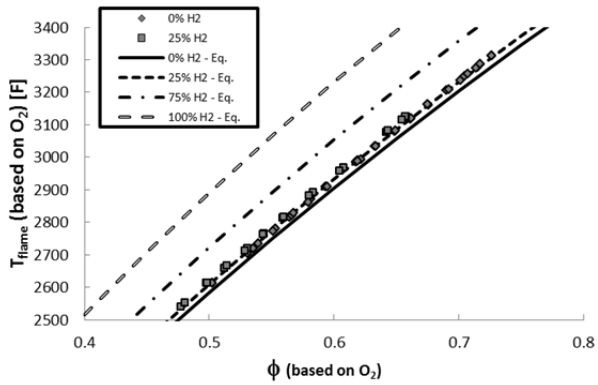


Figure 6. Impact of H2 addition on flame temperature

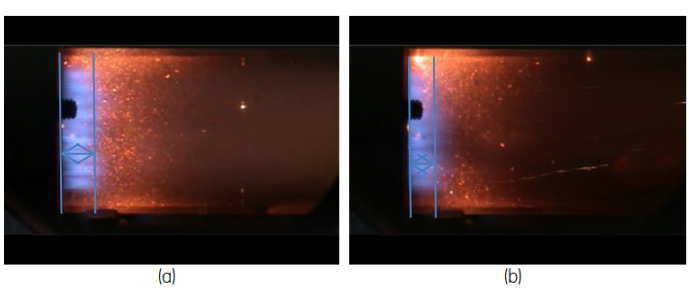
4.2 Impact of H2 doping on CO emissions and LBO.

Flame shape provides valuable information when it comes to understanding flame CO and LBO behavior. Figure 7 shows the flame structure for the natural gas flame (left) and the hydrogen enriched natural gas flame with 25% H2 addition (right). The flame resides slightly closer to the burner with hydrogen due to the increase in flame speed. The most notable difference is that the hydrogen enriched flame appears blurrier than the natural gas flame due to the dilution of the CH radical. The overall flame structure does not change dramatically for the hydrogen amounts considered in this study. At the highest hydrogen volume fractions used some flashback was experienced at high temperatures where the flame speed is highest.

Figure 8 presents the effect of H2 enrichment on CO emissions. There is no effect on the CO emissions with hydrogen enrichment until the LBO limit is approached. Up to this point the main generation mechanism for CO is dissociation which is driven by Tflame and hence no effect on CO in this regime. However, the story is different once LBO is approached.

Figure 8 shows that at a temperature of 2640 F, addition of H2 reduces the CO concentration from 500 ppm to single digit ppm even if the H2 percentage is as low as 5%. Figure 9 shows the flame shape with and without H2 addition under such conditions. Operating under this regime, close to the LBO limit, the flame temperature is low and most of the oxidation reactions rates decrease significantly. However, the increase in OH concentration that associates H2 addition promotes oxidation reactions (CO+OH=CO2+H) and CO is oxidized to CO2. This explanation is supported by flame images given in Figure 9. It is clear that flame stability is significantly improved (anchoring toward stabilizer) and flame length is also shorter with H2 addition.

For design purposes, it is important to translate that behavior in terms of impact on LBO. Figure 10 shows the impact of H2 addition on LBO. The curve shows a straight line relationship for the range of H2 concentrations used in this study. On average there is ~ 7% decrease in LBO temperature per percent H2 added.



(a) (b)

Figure 7. Flame luminosity images (a) NG/Air flame and (b) NG+25%H2/Air flame at 3250 F.

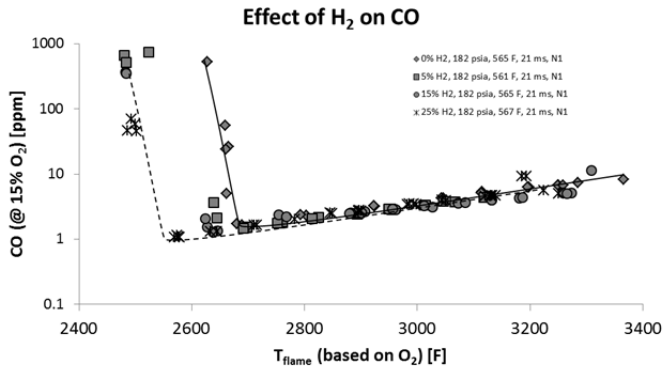
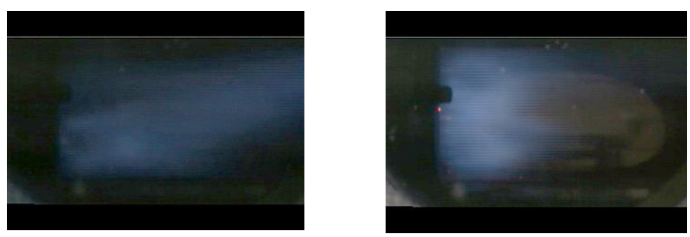


Figure 8. Impact of hydrogen addition on CO emissions (corrected to 15% oxygen).



(a) (b)

Figure 9. Flame luminosity near LBO (a) NG/Air flame and (b) NG+H2/Air flame.

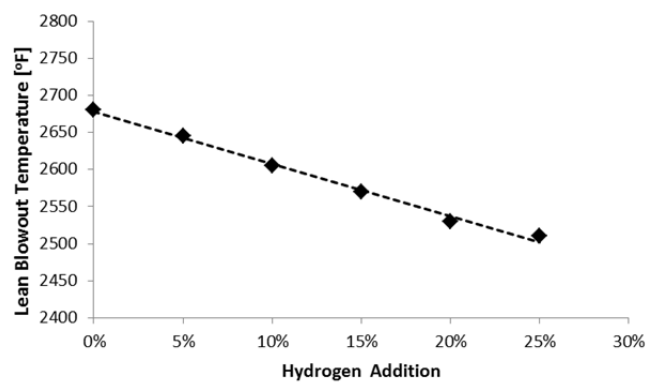


Figure 10. Impact of H2 addition on LBO.

4.3 Chemical Kinetics modelling

A chemical network reacting model was constructed to be utilized to understand the impact of H2 enrichment on NOx generation. The simplified model is shown in Figure 11. The network consists of an inlet, a Perfectly Stirred Reactor (PSR) and a Plug Flow Reactor (PFR). A major simplification in the model is possible since the air and fuel are perfectly premixed before being introduced to the nozzle. Table 1 shows a summary of the input to the reactor network. The reactor network was run for all cases and all flame temperatures of interest.

Figure 12 shows a comparison between the NOx emissions collected from the experiments and reactor network model. Only the data with no hydrogen (pure natural gas) is shown. Also, a correction of flame temperature due to heat loss is provided on the plot. It is clear that the model

is capable of predicting the NOx emissions as long as the flame temperature is defined properly. Elkady et. al did detailed analysis on this combustor and provided detailed heat loss analysis. However, both combustors are not identical as the data presented here is for 14 ms while the data they discussed was for 20 ms and did not include H2 enrichment data as we will discuss shortly.

Figure 13 shows the impact of H2 enrichment using the constructed reactor model. It is clear that there limited to no impact as we observed from the experimental data. However, for all the cases we ran the PSR residence time was fixed to 0.5 ms. While the experimental data does show that there is a limited impact but a close look at the data shows that there is a slight increase in the NOx emissions (around 0.5 ppm). The cost of running reactor network model to understand this problem is much less than running expensive experiments with laser diagnostics. Therefore, the reactor network model was further used to shed more light on this behavior.

Flame luminosity images (see Figure 7) suggest that there are differences in the flame size/location with and without H2 enrichment. Close look at the images shows that the flame is pulled back around 20%. This 20% corresponds to ~ ½'' or 1 ms. Therefore, we increase the PFR residence time by 0.5 ms and rerun the model. Figure 8 shows a comparison for the 20% H2 enrichment case with the standard 0.5 ms and extended residence time with 1 ms. The curve shifted with around 0.3 ppm in agreement with the slight increase that was observed form the experimental data.

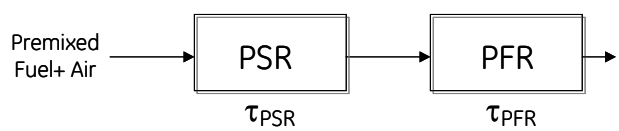


Figure 11. Chemical kinetic reactor network for emissions modelling

Table 1: Run conditions for chemical kinetics model

PARAMETER	VALUE	UNITS
Total residence time	14	ms
T3	633	K
P3	12.3	atm
Baseline CH4%	0.95486	By mass
Baseline C2H6%	0.037916	By mass
Baseline CO2%	0.001181	By mass
Baseline N2%	0.00434	By mass
H2%	0-0.04	By mass

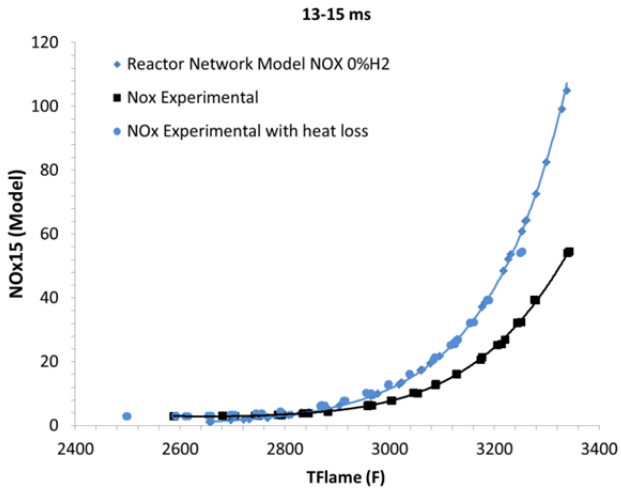


Figure 12. Comparison between NOx emissions measured from the experiments vs. modeled using chemical kinetics.

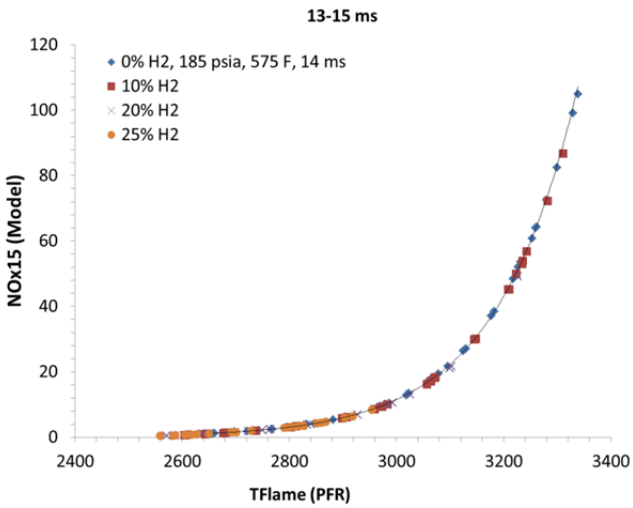


Figure 13. Impact of H2 enrichment using chemical kinetics (reactor model).

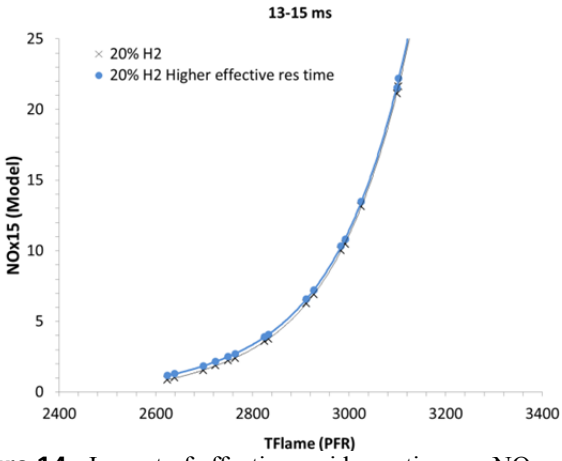


Figure 14. Impact of effective residence time on NOx emissions using reactor network modelling.

5 CONCLUSIONS

H2 enrichment up to 25% has limited impact on NOx emissions for perfectly premixed combustion. This was shown both from the experimental data that was collected under high pressure in a perfectly premixed combustor as well as from a reactor network model. The slight increase may be attributed to the change in flame shape and hence the change in effective residence time since H2 addition typically results in flame sitting closer to the fuel nozzle. Flame luminosity images show this behavior for the combustor under investigation. The discrepancy in published literature is due to comparison at non-identical flame temperature conditions. At a given equivalence ratio the flame temperature changes slightly with and without H2 addition only up to ~30% beyond which the change is dramatic. H2 enrichment has significant impact on extending the LBO limit. Data shows that the derivative is ~ 7% decrease in LBO temperature per percent H2 added.

6 REFRENCES

1. Griebel, P., E. Boschek, and P. Jansohn, *Lean Blowout Limits and NOx Emissions of Turbulent, Lean Premixed, Hydrogen-Enriched Methane/Air Flames at High Pressure.* Journal of Engineering for Gas Turbines and Power, 2007. **129**(2): p. 404.
2. Kim, H., V. Arghode, and A. Gupta, *Flame characteristics of hydrogen-enriched methane–air premixed swirling flames.* International Journal of Hydrogen Energy, 2009. **34**(2): p. 1063-1073.
3. Cozzi, F. and A. Coghe, *Behavior of hydrogen-enriched non-premixed swirled natural gas flames.* International Journal of Hydrogen Energy, 2006. **31**(6): p. 669-677.
4. Schefer and R., *Hydrogen enrichment for improved lean flame stability.* International Journal of Hydrogen Energy, 2003. **28**(10): p. 1131-1141.
5. Nguyen, O. and S. Samuelsen. *The Effect of Discrete Pilot Hydrogen Dopant Injection on the Lean Blowout Performance of a Model Gas Turbine Combustor*. in *The International Gas Turbine and Aeroengine Congress and Exhibition*. 1999. Indianapolis.
6. Jackson, G.S., et al., *Influence of H2 on the Response of Lean Premixed CH4 Flames to High Strained Flows.* Combustion and Flame, 2003. **132**(3): p. 503-511.
7. Tuncer, O., S. Acharya, and J. Uhm, *Dynamics, NOx and flashback characteristics of confined premixed hydrogen-enriched methane flames.* International Journal of Hydrogen Energy, 2009. **34**(1): p. 496-506.
8. Schefer, R.W. *Combustion Of Hydrogen-Enriched Methane In A Lean Premixed Swirl Burner*. in *Proceedings of the 2001 DOE Hydrogen Program Review*. 2001.
9. Wicksall, D., et al., *The interaction of flame and flow field in a lean premixed swirl-stabilized combustor operated on H/CH/air.* Proceedings of the Combustion Institute, 2005. **30**(2): p. 2875-2883.
10. Dagaut, P. and G. Dayma, *Hydrogen-enriched natural gas blend oxidation under high-pressure conditions: Experimental and detailed chemical kinetic modeling.* International Journal of Hydrogen Energy, 2006. **31**(4): p. 505-515.
11. Leonard, G. and S. Correa. *NOx Formation in Premixed High-Pressure Lean Methane Flames*. in *Fossil Fuel Combustion Symposium: 13th Annual Energy-Sources Technology Conference and Exhibition*. 1990. New Orleans, USA.
12. ElKady, A.M., et al., *Application of Exhaust Gas Recirculation in a DLN F-Class Combustion System for Postcombustion Carbon Capture.* Journal of Engineering for Gas Turbines and Power, 2009. **131**(3): p. 034505.
13. Evulet, A.T., et al., *On the Performance and Operability of GE's Dry Low NOx Combustors utilizing Exhaust Gas Recirculation for PostCombustion Carbon Capture.* Energy Procedia, 2009. **1**(1): p. 3809-3816.
14. Elkady, A.M., et al. *Gas Turbine Emission Characteristics In Perfectly Premixed Combustion*. in *Proceedings of ASME Turbo Expo 2011: Power for Land, Sea and Air*. 2011. Vancouver, Canada: GT2011-46470.
15. Ahmed, M.E., et al. *Exhaust Gas Recirculation in DLN F-Class Gas Turbines for Post-Combustion CO2 Capture*. in *ASME Turbo Expo 2008: Power for Land, Sea, and Air*. 2015: American Society of Mechanical Engineers.
16. Elkady, A.M., et al. *Exhaust Gas Recirculation Performance in Dry Low Emissions Combustors*. in *ASME 2011 Turbo Expo: Turbine Technical Conference and Exposition*. 2015: American Society of Mechanical Engineers.
17. Graham, G., et al. *Modeling CO With Flamelet-Generated Manifolds: Part 2 — Application*. in *ASME Turbo Expo 2012: Turbine Technical Conference and Exposition*. 2015: American Society of Mechanical Engineers.
18. Karim, G.G., et al. *Modeling CO With Flamelet-Generated Manifolds: Part 1—Flamelet Configuration*. in *ASME Turbo Expo 2012: Turbine Technical Conference and Exposition*. 2015: American Society of Mechanical Engineers.